

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1498

Pauline 69 kV
July 26, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1498. The Interconnection Customer proposes a 178.1 MW solar photovoltaic facility interconnecting at the Pauline 69 kV substation in the North Carolina area. The request was received on June 14, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 363 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$281,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Avra Valley – Windsor Dp 138 KV: 100.4% loading (439 MVA) following the outage of the Mira Vista – Yarnen 345 KV L; DFAX = 6.9%.

Windsor Dp – Werner West 138 KV: 100.2% loading (451 MVA) following the outage of the Mira Vista – Yarnen 345 KV L; DFAX = 6.9%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Avra Vly – Windsor Dp 138 KV line	Reconductor line	\$1,651,000	6% (\$92,000)
Werner West – Windsor Dp 138 KV	Reconductor line	\$3,389,000	6% (\$189,000)
	Total allocated to project		\$281,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.